User Intent Classification for Hallucination Reduction in LLMs

Review 2 — Literature Review & Architecture

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Literature Review — Hallucination Detection

Title (Year & Authors)	Journal / Venue	Approach & Technology	Key Results / Performance	Limitations & Gaps
Maynez et al. (2020) Summarizing facts, faithfulness…	ACL 2020	Hallucination taxonomy; faithfulness vs. factuality distinction in abstractive summarization	Faithful summaries score ~84% by human eval; faithfulness & factuality often conflate	Limited to summarization; does not generalize to QA or conversational LLMs
Shi et al. (2023) Large Language Models Easily Distracted by Irrelevant Context	ICML 2023	Empirical study; GPT-3/4 on grade-school math with injected irrelevant sentences	Accuracy drops by up to 20% when irrelevant context injected	Narrow to math; no routing or mitigation strategy proposed
Manakul et al. (2023) SelfCheckGPT	EMNLP 2023	Consistency-based hallucination detection via multiple stochastic samples; NLI scoring	Outperforms perplexity-based baselines; AUC ~0.82 on WikiBio	Post-hoc only; adds 3–5× inference cost; does not prevent generation
Kadavath et al. (2022) Language Models (Mostly) Know What They Know	arXiv 2022	Self-evaluation of own uncertainty via P(True); tested on GPT-like models	Calibration improves with scale; models reliably flag own uncertainty at 50B+ params	Requires very large model; confidence estimates degrade on adversarial queries
Farquhar et al. (2024) Detecting Hallucinations with Semantic Entropy	Nature 2024	Clusters semantically equivalent generations; flags high-entropy clusters as hallucinations	SotA on hallucination detection; F1 ~0.79 on TriviaQA	High compute cost (multi-sample); no integration with routing; detects, does not prevent

Literature Review — Intent & Query Classification

Title (Year & Authors)	Journal / Venue	Approach & Technology	Key Results / Performance	Limitations & Gaps
Chen et al. (2021) Intent Detection with BERT	ACL 2021	Fine-tuned BERT on SNIPS/ATIS datasets for user intent classification	98.6% accuracy on SNIPS; 97.5% on ATIS	Closed-domain intents only; no mapping to tool-routing or hallucination prevention
Liu et al. (2019) RoBERTa for Text Classification	NAACL 2019	Pre-training improvements; evaluated on GLUE, intent benchmarks	~1% gain over BERT; strong baseline for classification tasks	Classification task only; no tool-call routing studied; no cost analysis
Larson et al. (2019) Evaluation Dataset for Intent Detection OOS	EMNLP 2019	Introduces CLINC-150 dataset; OOS (out-of-scope) query detection	96.9% in-scope accuracy; OOS detection degrades to ~75%	Binary in-scope vs. OOS framing; does not distinguish fact/math/creative categories
Ouyang et al. (2022) InstructGPT (RLHF)	NeurIPS 2022	RLHF to align LLM responses with user intent; instruction-following fine-tuning	Preferred 85% of the time over GPT-3 (human eval)	Intent alignment ≠ intent classification; no explicit routing; high training cost

Literature Review — Tool-Augmented Generation

Title (Year & Authors)	Journal / Venue	Approach & Technology	Key Results / Performance	Limitations & Gaps
Schick et al. (2023) Toolformer	NeurIPS 2023	Self-supervised insertion of API calls (calculator, search, calendar) into generated text	Outperforms GPT-3 on math/QA; competitive with GPT-3.5 on some tasks	Tool selection is implicit; no isolated routing benchmark; cold-start API calls
Gao et al. (2023) PAL — Program-Aided Language Models	ICML 2023	Offloads all arithmetic to a Python interpreter; LLM produces code, not answers	Accuracy +10–20% over direct prompting on GSM8K, MATH	Restricted to computational queries; routing to PAL assumed, not learned
Yao et al. (2023) ReAct: Synergizing Reasoning and Acting	ICLR 2023	Interleaves chain-of-thought reasoning with tool calls; tested on HotpotQA, FEVER	Outperforms chain-of-thought alone; 73.3% on HotpotQA	Routing baked into prompt structure; no lightweight classifier; high token usage
Trinh et al. (2024) AlphaGeometry	Nature 2024	LLM proposes language constructs; symbolic solver derives proofs for olympiad geometry	Solves 25/30 IMO geometry problems; comparable to gold medalists	Domain-specific (geometry only); routing not generalizable; not a practical classifier

Literature Review — Retrieval-Augmented Generation & Model Routing

Title (Year & Authors)	Journal / Venue	Approach & Technology	Key Results / Performance	Limitations & Gaps
Lewis et al. (2020) **RAG — Retrieval-Augmented Generation**	NeurIPS 2020	Combines dense retrieval (DPR) with seq2seq generation; trained end-to-end	SotA on Natural Questions, TriviaQA; F1 +5–8% over closed-book GPT	Retrieval quality limits generation; no intent classifier; high latency
Mallen et al. (2023) **When Not to Trust Language Models**	ACL 2023	Popularity-based retrieval trigger; selectively queries Wikipedia for rare entities	+4% F1 on EntityQuestions; reduces unnecessary retrieval by ~60%	Popularity heuristic only; no generalized classifier; does not cover math queries
Shnitzer et al. (2023) **Large Language Model Routing with Benchmark Datasets**	arXiv 2023	Learns to route queries between large and small LLMs based on predicted accuracy	~90% of GPT-4 accuracy at ~40% cost using routing	Routing between LLMs, not to tools; no intent categories; accuracy gap ignored
Hu et al. (2022) **LoRA — Low-Rank Adaptation**	ICLR 2022	Parameter-efficient fine-tuning via low-rank matrix injection into transformer layers	Matches full fine-tune on GLUE at ~0.1% trainable params; GPU memory ×3 reduction	Adapter technique only; no routing or hallucination study; not a classifier by itself

Literature Review — Benchmarks & Evaluation Frameworks

Title (Year & Authors)	Journal / Venue	Approach & Technology	Key Results / Performance	Limitations & Gaps
Hendrycks et al. (2021) Measuring Multitask Language Understanding (MMLU)	ICLR 2021	57-subject MCQ benchmark spanning STEM, humanities, social sciences; zero-shot/few-shot	GPT-4: 86.4%; human expert: ~89.8% on full MMLU	MCQ format; no routing signal; does not differentiate factual vs. computational need
Cobbe et al. (2021) GSM8K — Grade School Math Benchmark	arXiv 2021	8,500 multi-step arithmetic problems; tests step-by-step reasoning ability	GPT-4 (CoT): 92%; prior SotA (fine-tuned): ~35%	Math only; all queries route to calculator — no mixed-intent benchmark
Joshi et al. (2017) TriviaQA	ACL 2017	95K trivia QA pairs with evidence documents; tests factual retrieval	Human: 79.7% F1; SotA models: ~88% F1	Fact-lookup only; no routing or multi-intent evaluation
Liang et al. (2022) Holistic Evaluation of Language Models (HELM)	arXiv 2022	Multi-metric LLM benchmark: accuracy, calibration, robustness, fairness, efficiency	Comprehensive across 42 scenarios and 30+ metrics	Evaluates model outputs, not routing decisions; no cost-accuracy tradeoff for routing

Summary of Research Gaps

01

Post-hoc hallucination detection only

Semantic entropy, SelfCheckGPT, and NLI verification detect hallucinations after generation. None prevent the hallucination-prone call from being made.

02

Routing is implicit inside large models

ChatGPT, Gemini, and Claude route internally via full-scale LLM inference. No prior work isolates this routing decision as a separate, benchmarked component.

03

Intent classifiers not connected to tool-routing

BERT/RoBERTa intent classifiers (SNIPS, ATIS, CLINC) classify dialogue acts — not whether a query needs a calculator, search engine, or open-ended generation.

04

No cost-accuracy tradeoff reported for routing

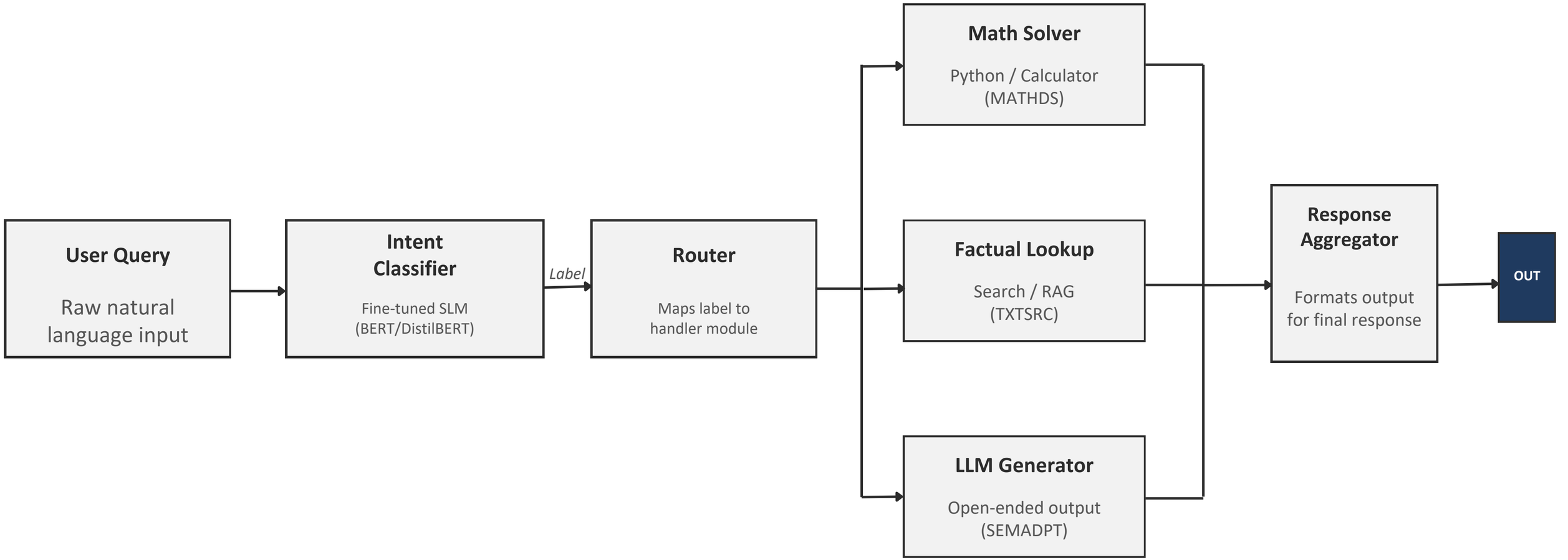
Routing studies compare accuracy but omit latency and inference cost. No study reports: how much accuracy is lost, and how much cost is saved, under matched conditions.

Proposed Methodology

Our Approach

1. Classify user intent before generating a response.
2. Use a separate module to route the query to the right category.
3. Group intents into 4 classes: MATHDS, TXTSRC, SEMADPT, CSTRSET.
4. Fine-tune a small language model to act as the intent classifier.
5. Compare the classifier with a large LLM (zero-shot) on the same test set.
6. Report both accuracy and inference cost for a fair comparison.

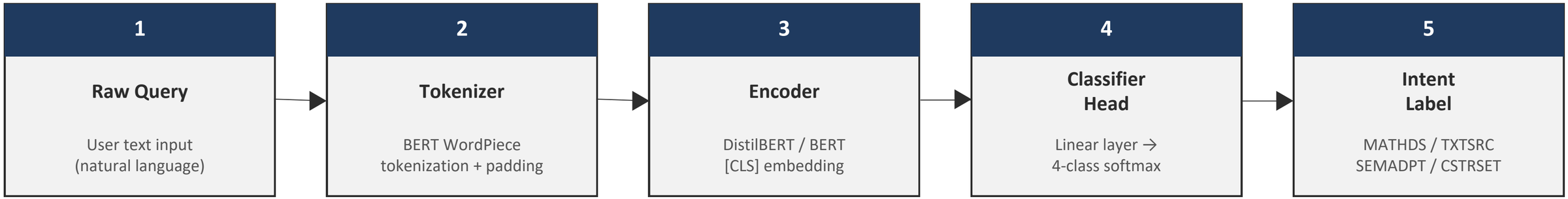
Complete System Architecture



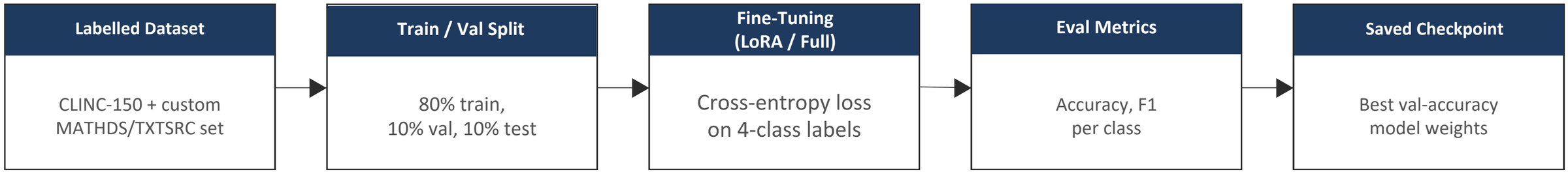
Note: The Intent Classifier is the core module under study. The LLM Baseline replaces it with a zero-shot large LLM call — both paths are evaluated on the same test set for accuracy and cost.

Data Flow Diagram — Intent Classifier Module

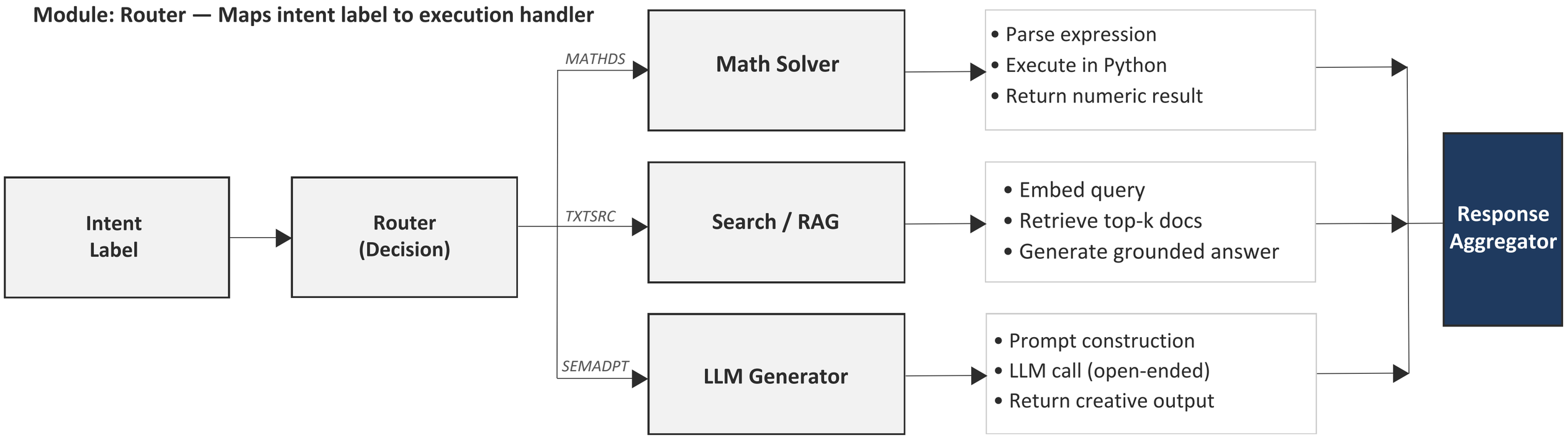
Module: Intent Classifier (SLM Fine-Tuning Pipeline)



Training Data Flow

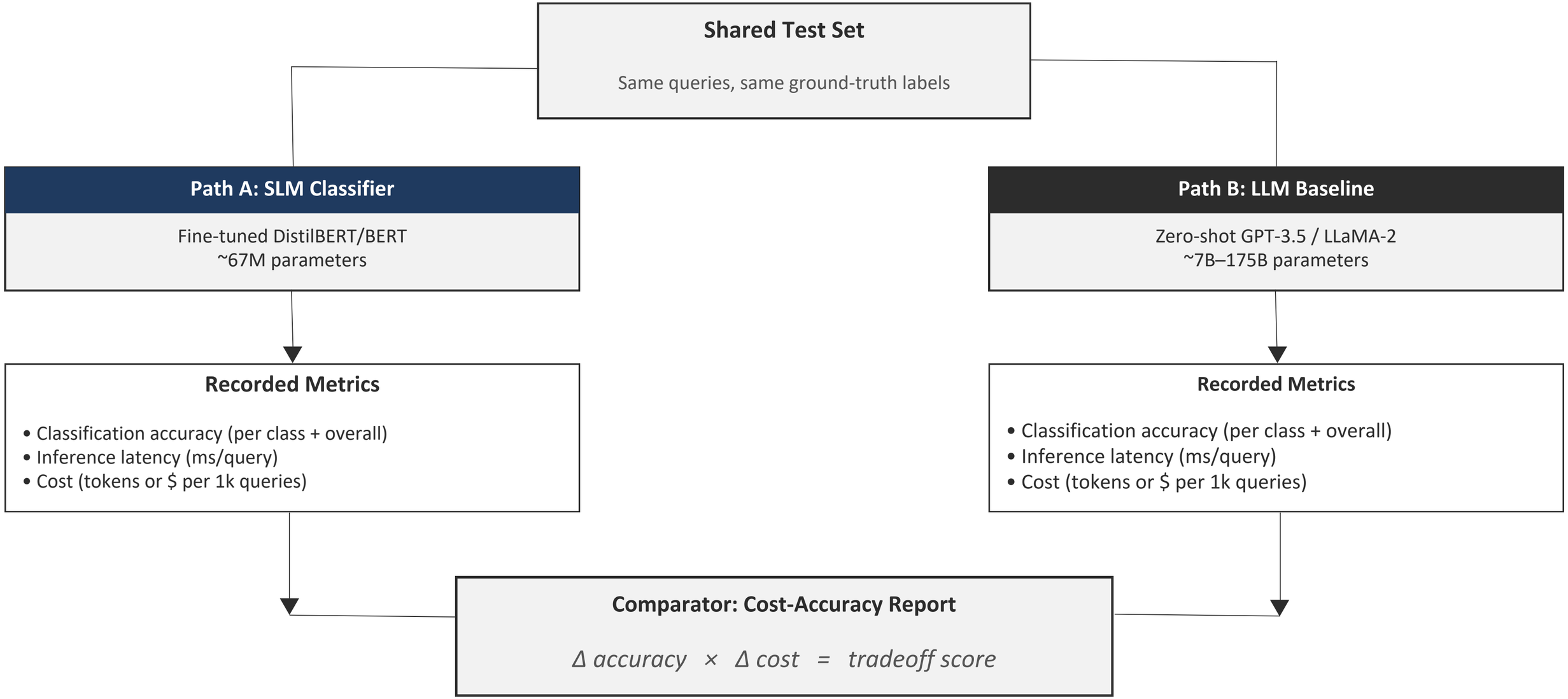


Data Flow Diagram — Router & Tool Execution Modules



Data Flow Diagram — Evaluation & Comparison Pipeline

How SLM Classifier and LLM Baseline are compared under identical conditions



Thank You!

